INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

MODULE-3

GUEST ANALYSIS MODULE

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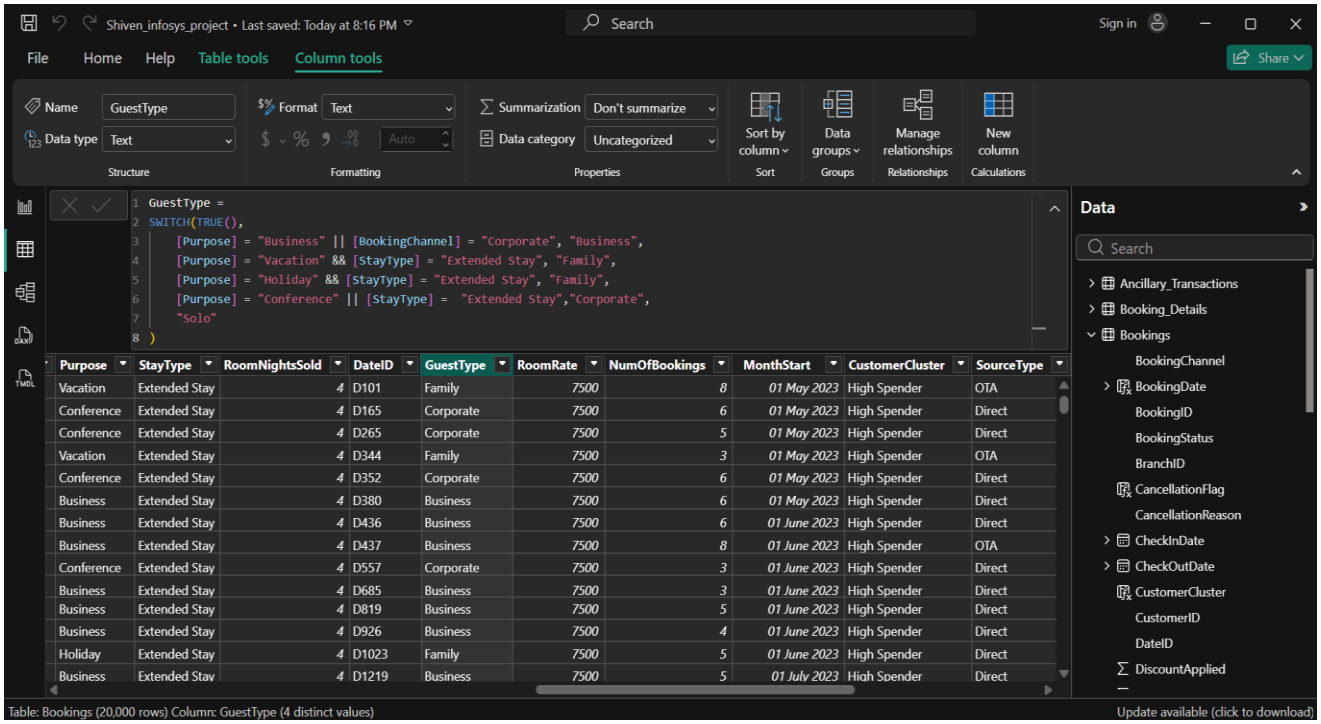
Project Name: HotelRevAI AI Driven Revenue Analysis for Hotels_Batch -13

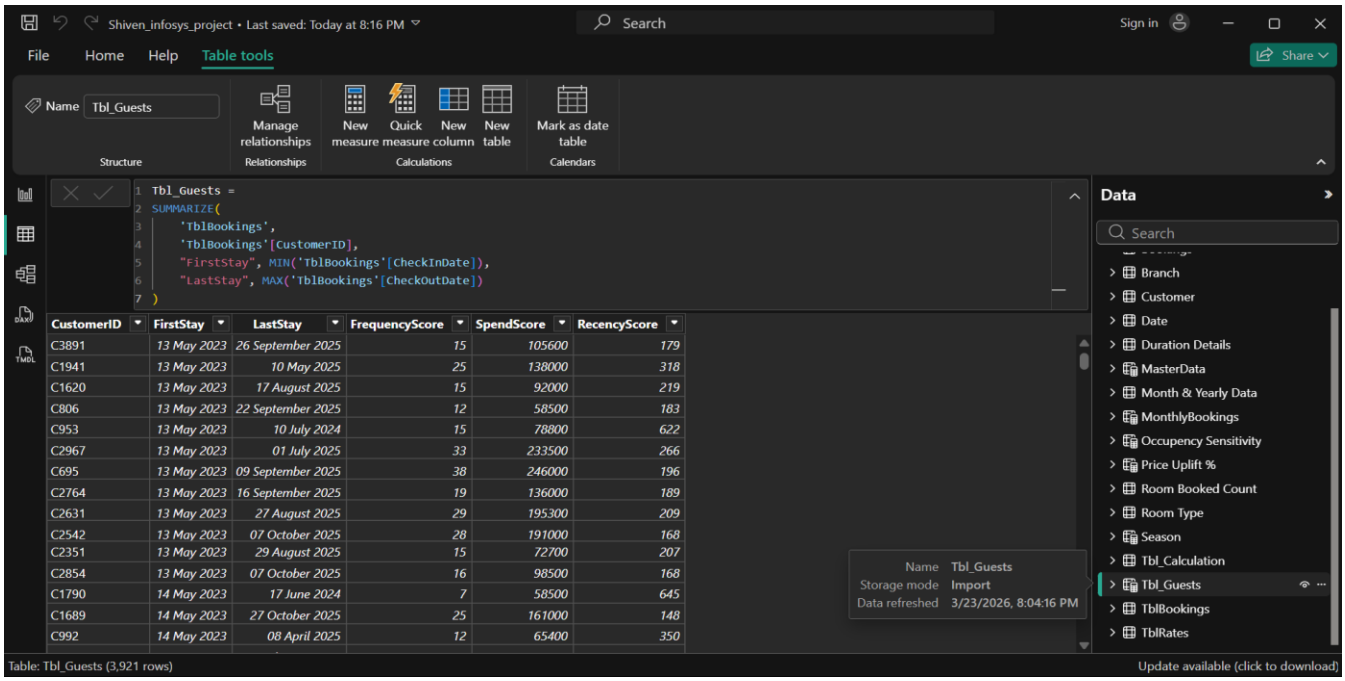
1)After refreshing and loading the remaining data in Power BI, several additional columns were introduced in the Booking table to enrich the analysis. These included *Source Type*, *Guest Type*, *Room Rate*, *Number of Bookings*, and *Customer Cluster*.

Adding these fields provides deeper insights into booking behavior, customer segmentation, and revenue patterns, making the analysis more detailed and meaningful.



```
SourceType =
SWITCH(
   'Bookings'[BookingChannel],
   "Call Center", "Direct",
   "Mobile App", "Direct",
   "Website", "Direct",
   "Travel Agent", "OTA",
   "Other"
)

GuestType =
SWITCH(TRUE(),
   [Purpose] = "Business" || [BookingChannel] = "Corporate", "Business",
   [Purpose] = "Vacation" && [StayType] = "Extended Stay", "Family",
   [Purpose] = "Holiday" && [StayType] = "Extended Stay", "Family",
   [Purpose] = "Conference" || [StayType] =  "Extended Stay","Corporate",
   "Solo"
)

RoomRate = RELATED(Room[Price])

NumOfBookings =
CALCULATE(
   COUNT(Bookings[BookingID]),
   ALLEXCEPT(Bookings, Bookings[CustomerID])
)

CustomerCluster =
SWITCH(TRUE(),
   [NumOfBookings] = 1 && [Revenue] < 5000, "First-Timer",
   [NumOfBookings] >= 3 && [Revenue] < 15000, "Loyal Guest",
   [Revenue] >= 15000, "High Spender",
   "Other"
)
```

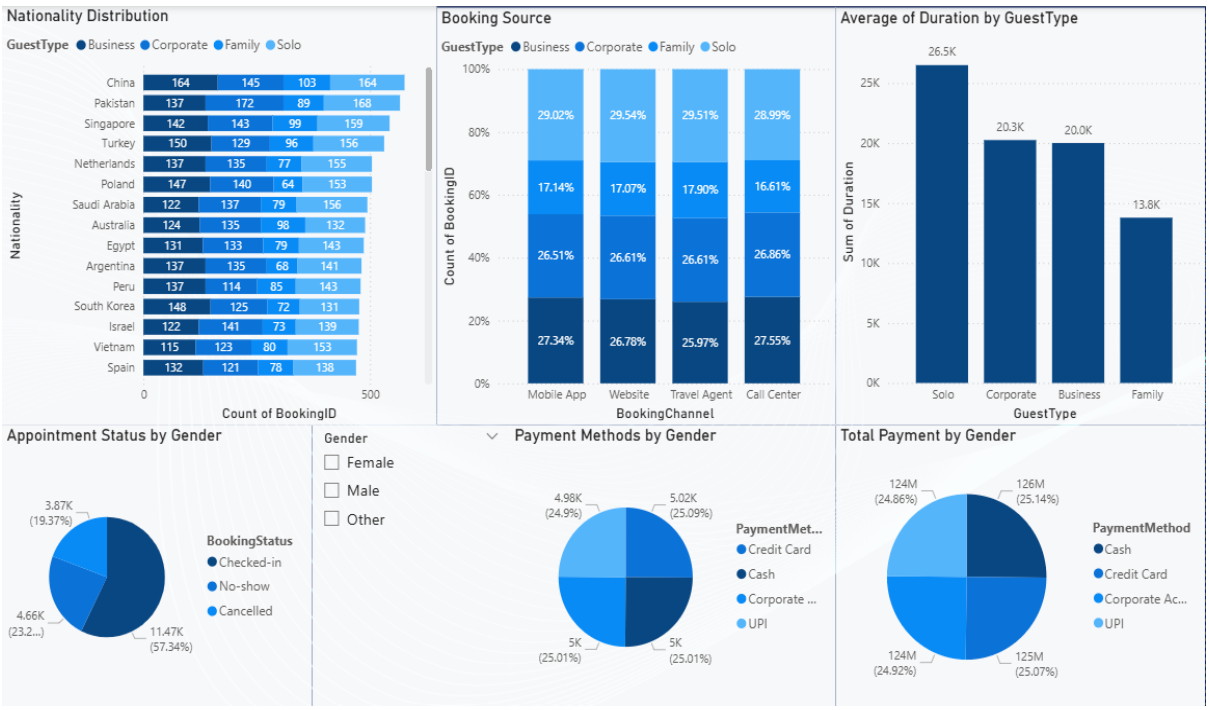
2) After that, a new table called TblGuests was created in Power BI. In this table, key scoring columns—*Frequency Score*, *Spend Score*, and *Recency Score*—were added.
These metrics help in evaluating customer behavior by analyzing how often guests book, how much they spend, and how recently they made a booking, enabling better segmentation and insights.



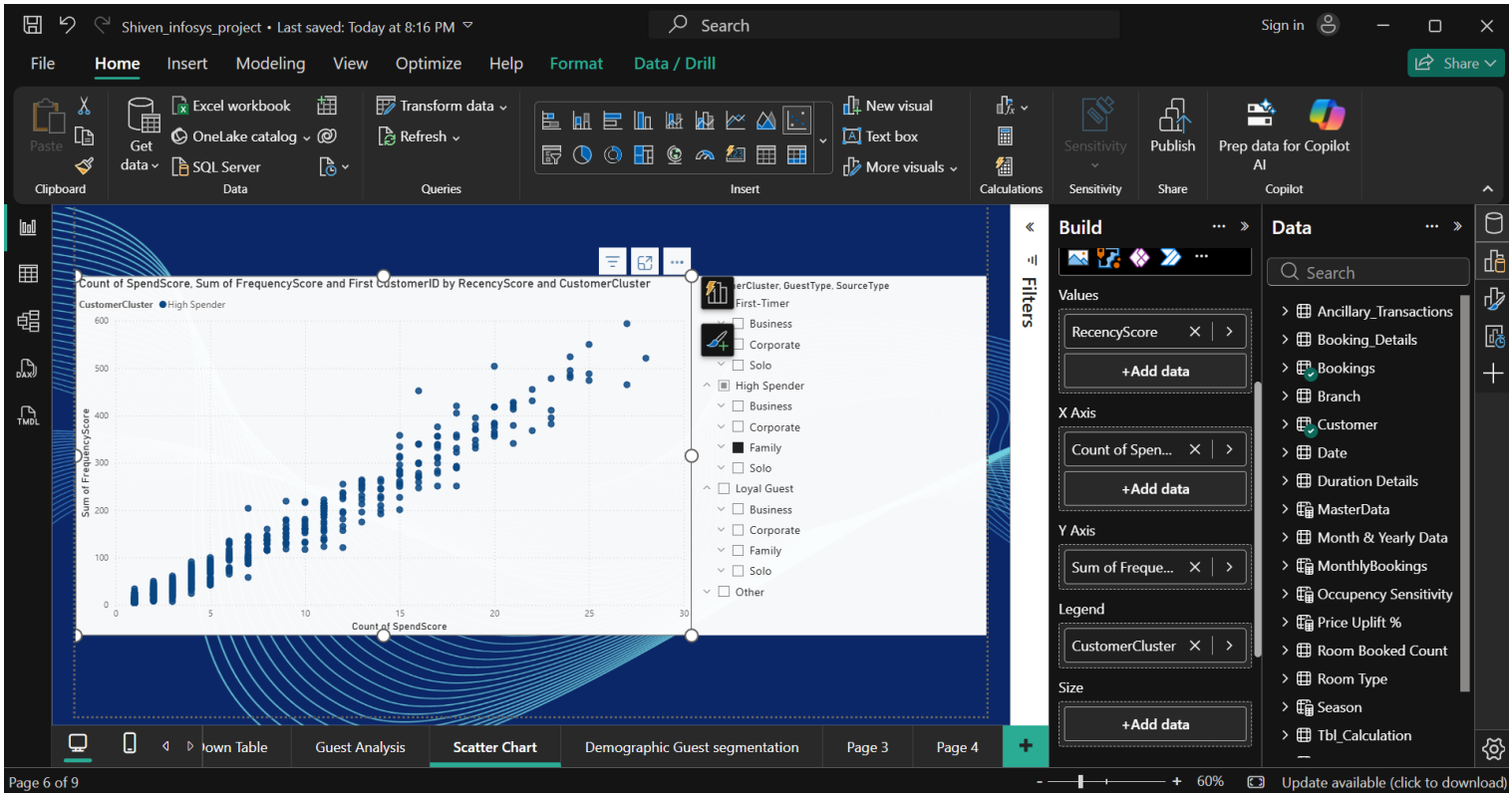
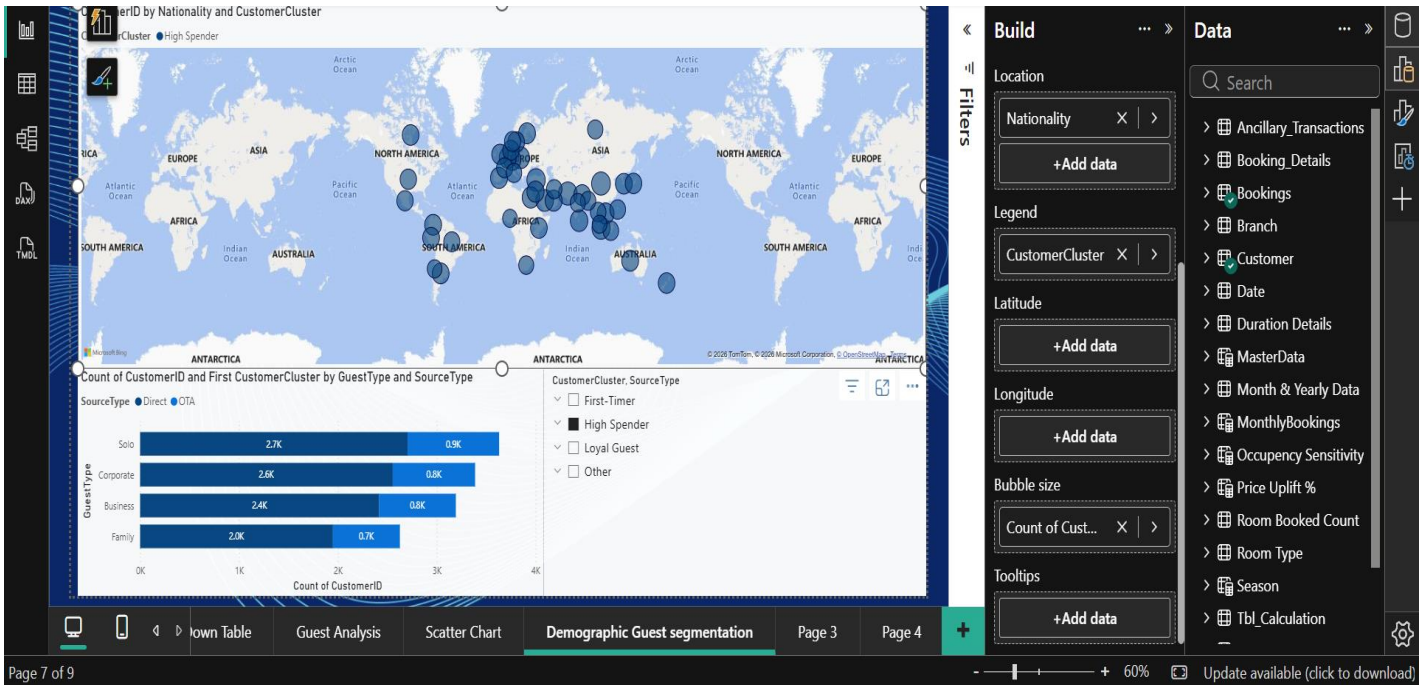
CustomerID	FirstStay	LastStay	FrequencyScore	SpendScore	RecencyScore
C3891	13 May 2023	26 September 2025	15	105600	179
C1941	13 May 2023	10 May 2025	25	138000	318
C1620	13 May 2023	17 August 2025	15	92000	219
C806	13 May 2023	22 September 2025	12	58500	183
C953	13 May 2023	10 July 2024	15	78800	622
C2967	13 May 2023	01 July 2025	33	233500	266
C695	13 May 2023	09 September 2025	38	246000	196
C2764	13 May 2023	16 September 2025	19	136000	189
C2631	13 May 2023	27 August 2025	29	195300	209
C2542	13 May 2023	07 October 2025	28	191000	168
C2351	13 May 2023	29 August 2025	15	72700	207
C2854	13 May 2023	07 October 2025	16	98500	168
C1790	14 May 2023	17 June 2024	7	58500	645
C1689	14 May 2023	27 October 2025	25	161000	148
C992	14 May 2023	08 April 2025	12	65400	350

VISUALISATION

Guest Analysis:



Scatter Chart (Based on sum of SpendScore, sum of FrequencyScore and CustomerID by RecencyScore and CustomerCluster):



Demographic Guest Segmentation (Based on Nationality , CustomerCluster, SourceType, GuestType):